copy into the file and save. As mentioned, the logging is split into two filters; one of which runs before each operation, and one which runs afterwards. This ensures that if the process crashes while processing a request we can still see what requests where being processed at the time. If we only logged responses, then we could lose any trace of a request if the process crashes. To allow somebody reviewing the logs to correlate requests with responses we generate a unique audit log id in the before() filter and add it as an attribute to the request. In the after() filter we can then retrieve the same audit log id so that the two log events can be tied together.

For simplicity in this example, we provide an unauthenticated REST endpoint for retrieving recent audit log entries. Normally audit logs should never be exposed in this way as they may contain sensitive or private details.

Listing 3.5 The audit log controller

```
package com.manning.apisecurityinaction.controller;

import org.dalesbred.*;
import org.json.*;
import spark.*;

import java.time.*;
import java.time.temporal.*;
import java.util.stream.*;

public class AuditController {

  private final Database database;
```